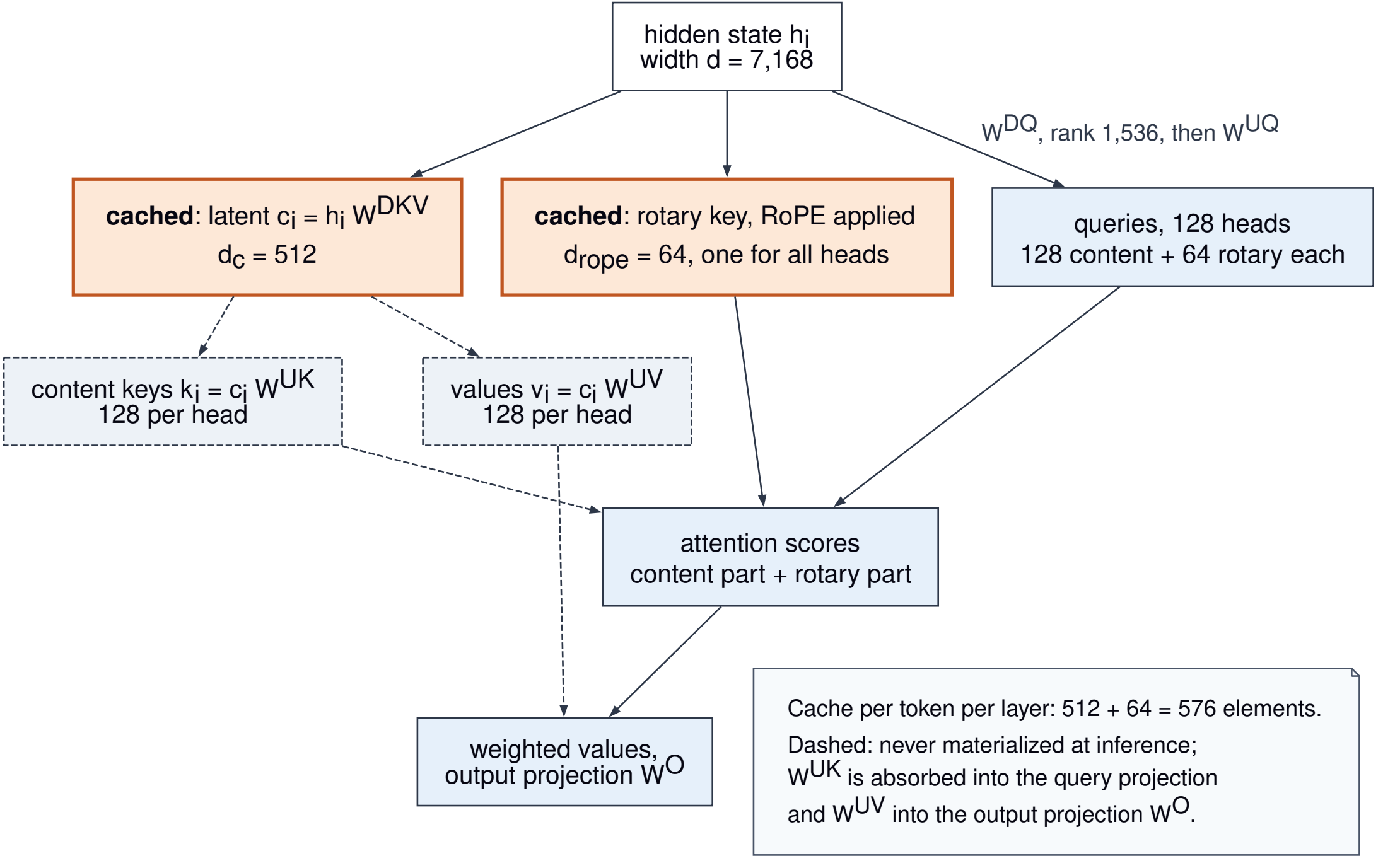
hidden state h_i
width d = 7,168
W^{DQ}, rank 1,536, then W^{UQ}
cached: latent $c_i = h_i W^{DKV}$
d_c = 512
cached: rotary key, RoPE applied
d_{rope} = 64, one for all heads
queries, 128 heads
128 content + 64 rotary each
content keys $k_i = c_i W^{UK}$
128 per head
values $v_i = c_i W^{UV}$
128 per head
attention scores
content part + rotary part
weighted values,
output projection W^O
Cache per token per layer: 512 + 64 = 576 elements.
Dashed: never materialized at inference;
W^{UK} is absorbed into the query projection
and W^{UV} into the output projection W^O.